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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
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Dear Prof. Merz,

We submit our manuscript **“Resistance-Aware Docking Prioritization and Short MD Stress Tests Measure Non-Equivalent Quantities in an Antimalarial Candidate Cohort”** for consideration as an Article in *JCIM*.

The central question is whether a resistance-aware, target-level docking workflow (RRS + PNS + ACSI) produces a coherent prioritization signal inside a curated antimalarial cohort, and whether a short single-replicate MD structural-stress test is concordant with that signal or instead reveals non-equivalence. Seventeen Set-C candidates were ranked by a docking-derived resistance-retention score across six PfDHFR and PfCRT resistance states; chemical-space (ACSI) and network (PNS) scores were kept as separate ranking dimensions. On the 12 candidates with complete two-target data the classes were one A*, one A, four B, five C, and one D. Four parent-study complexes outside the Set-C list were simulated for 10 ns each as a pose-retention check; only PfCRT-214 yielded an interpretable MM-GBSA estimate (-18.25 ± 0.40 kcal mol⁻¹). A 16-system single-replicate pilot on two Set-C candidates did not recover the docking retention pattern (directional disagreement in seven of eight mutant comparisons that possessed a wild-type docking reference). An external 39-ligand panel averaged near RRS 100. The ranking is informative inside the computational panel only as a hypothesis-generating prioritization, not as a validated triage or filtering procedure; the short MD stress test demonstrates that the two quantities are non-equivalent. Neither result claims biological target engagement, affinity, or resistance circumvention.

The MD component is reported in accordance with the JCIM guidelines of Soares et al.: single-replicate limitation, force-field versions, ensemble, trajectory acceptance criteria, and data-availability statements are explicit. Companion Paper 1 supplies only upstream chemical-space context, not independent replication.

No external funding was received for this work. Code, source data, and analysis scripts are available on GitHub under an open-source licence; a Zenodo snapshot will be deposited at publication. Statistics are reported with effect sizes, permutation *p*-values, bootstrap intervals, and Bonferroni adjustment.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African natural-product computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free-energy methods and MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)